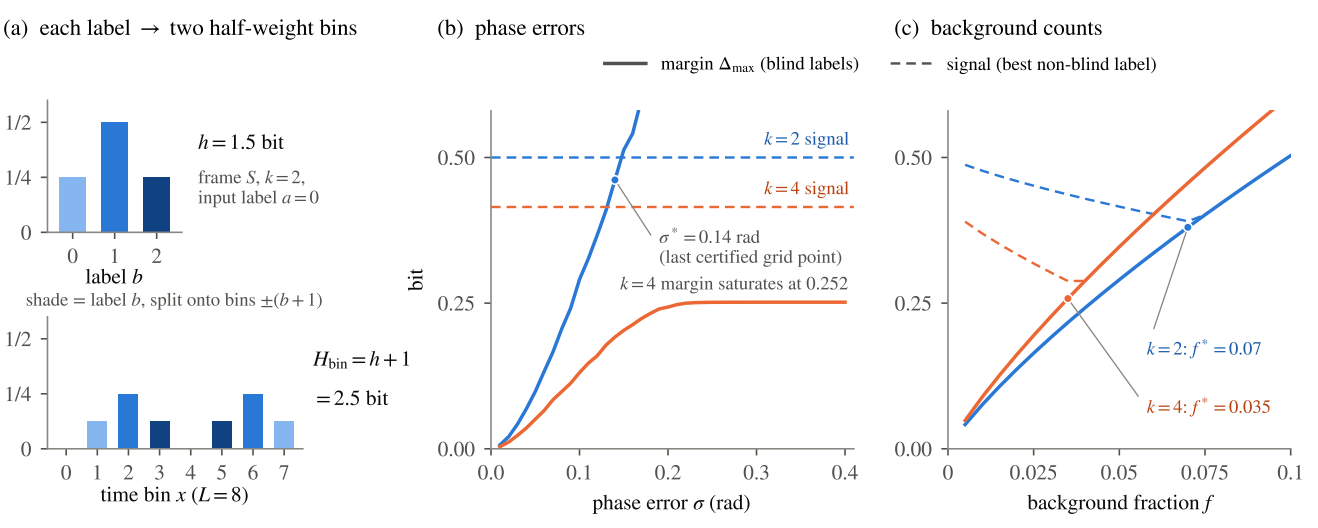
(a) each label → two half-weight bins
1/2
1/4
0
h = 1.5 bit
frame S, k = 2,
input label a = 0
0 1 2
label b
shade = label b, split onto bins ±(b + 1)
1/2
1/4
0
H_bin = h + 1
= 2.5 bit
0 1 2 3 4 5 6 7
time bin x (L = 8)
(b) phase errors
margin Δ_max (blind labels)
k = 2 signal
k = 4 signal
σ* = 0.14 rad
(last certified grid point)
k = 4 margin saturates at 0.252
bit
0.50
0.25
0.00
0.0 0.1 0.2 0.3 0.4
phase error σ (rad)
(c) background counts
signal (best non-blind label)
0.50
0.25
0.00
0 0.025 0.05 0.075 0.1
background fraction f
k = 2: f* = 0.07
k = 4: f* = 0.035